What is SQL?

SQL (Structured Query Language) is a language used to communicate with databases. It is used to store, retrieve, update, and delete data.

What are the types of SQL commands?

DDL (Data Definition Language)

Used to define database structure.

- CREATE
- ALTER
- DROP
- TRUNCATE

DML (Data Manipulation Language)

Used to manipulate data.

- INSERT
- UPDATE
- DELETE

DQL (Data Query Language)

Used to fetch data.

- SELECT

DCL (Data Control Language)

Used for permissions.

- GRANT
- REVOKE

TCL (Transaction Control Language)

Used for transaction management.

- COMMIT
- ROLLBACK
- SAVEPOINT

What is GRANT?

`GRANT` is used to give permissions to users on database objects like tables.

Syntax

```
GRANT permission
ON table_name
TO user_name;
```

Example

```
GRANT SELECT
ON student
TO user1;
```

Explanation

- `SELECT` permission is given
- `user1` can view data from `student` table

What is REVOKE?

`REVOKE` is used to remove permissions from users.

Syntax

```
REVOKE permission
ON table_name
FROM user_name;
```

Example

```
REVOKE SELECT
ON student
FROM user1;
```

Explanation

- Removes `SELECT` permission
- `user1` can no longer view data

Easy Interview Definition

GRANT

"GRANT is used to provide permissions to users in a database."

REVOKE

"REVOKE is used to remove permissions from users in a database."

COMMIT

Definition

`COMMIT` is used to permanently save changes in the database.

Example

```
UPDATE student
SET marks = 90
WHERE id = 1;

COMMIT;
```

Explanation

- Changes are saved permanently
- Cannot undo after commit

Interview Definition

"COMMIT permanently saves transaction changes into the database."

---

2. ROLLBACK

Definition

`ROLLBACK` is used to undo changes before commit.

Example

```
UPDATE student
SET marks = 90
WHERE id = 1;

ROLLBACK;
```

Explanation

- Cancels changes
- Database returns to previous state

Interview Definition

"ROLLBACK is used to undo transaction changes."

SAVEPOINT in SQL

Simple Definition

`SAVEPOINT` is used to create a temporary checkpoint inside a transaction.

If any mistake happens after that point, we can return back to the savepoint instead of cancelling the whole transaction.

Example

```
UPDATE student
SET marks = 90
WHERE id = 1;

SAVEPOINT s1;

DELETE FROM student
WHERE id = 2;

ROLLBACK TO s1;
```

Difference between DELETE, DROP, and TRUNCATE

DELETE	TRUNCATE	DROP
Deletes selected rows	Deletes all rows	Deletes entire table
WHERE clause can be used	WHERE cannot be used	Removes structure also
Can rollback	Cannot rollback mostly	Cannot rollback
Slower	Faster	Fastest

What is a Table?

A table is a collection of rows and columns used to store data.

What is a Primary Key?

Primary Key uniquely identifies each row in a table.

Features:

- Unique
- Cannot contain NULL
- Only one primary key per table

Foreign key

“A foreign key in one table refers to the primary key of another table, and it is used to establish the relationship between tables.”

A foreign key can be NULL and it can also contain duplicate values because it is used only to establish relationships between tables.”

Difference Between Primary Key and Foreign Key

Primary Key	Foreign Key
Unique identifier	Creates relation
No NULL values	NULL allowed
One per table	Multiple allowed

What is Normalization?

Normalization is the process of organizing data to reduce redundancy and improve efficiency.

Types:

- 1NF
- 2NF
- 3NF
- BCNF

1NF:

"1NF or First Normal Form means each column in a table should contain only single values and there should be no repeating groups.

Not in 1NF

Student_ID	Name	Subjects
101	Ram	Java, SQL

❌ Problem:

- `Subjects` column contains multiple values

Converted to 1NF

Student_ID	Name	Subject
101	Ram	Java
101	Ram	SQL

✅ Now:

- Each column contains only one value
- Table follows 1NF

2nf

“2NF means the table should be in 1NF and non-key attributes should depend on the complete primary key not partially.

2NF Simple Example

Table

Student_ID	Course	Student_Name
101	Java	Ram
101	SQL	Ram
102	Python	Ravi

Here:

- One student can join many courses
- So primary key becomes:

```
(Student_ID + Course)
```

Now see carefully:

- `Student_Name` depends only on `Student_ID`
- It does not depend on complete key `(Student_ID + Course)`

Because even if course changes,
student name remains same.

❌ This is partial dependency
❌ So table is not in 2NF

Convert into 2NF

Student Table

Student_ID	Student_Name
101	Ram

Student_ID	Student_Name
102	Ravi

Course Table

Student_ID	Course
101	Java
101	SQL
102	Python

✅ Partial dependency removed
✅ Table follows 2NF

3nf

3NF means the table should be in 2NF and non-key attributes should not depend on other non-key attributes.”

Table

Roll_No	Name	Department	HOD
1	Ram	CSE	Kumar
2	Ravi	ECE	Suresh

Now see:

- `Roll_No` decides student details
- But `HOD` actually depends on `Department`
- Not directly on `Roll_No`

Because:

```
Department → HOD
```

If department is CSE, HOD is always Kumar.

So:

- One non-key column (`HOD`)

- depends on another non-key column (`Department`)

❌ This is transitive dependency
❌ Not in 3NF

Convert into 3NF

Student Table

Roll_No	Name	Department
1	Ram	CSE
2	Ravi	ECE

Department Table

Department	HOD
CSE	Kumar
ECE	Suresh

✅ Now no unnecessary dependency
✅ This is 3NF

What is the difference between CHAR and VARCHAR?

CHAR	VARCHAR
Fixed length	Variable length
Faster	Saves memory
Wastes space	Efficient

What is a Subquery?

A query inside another query is called a subquery.

Difference between WHERE and HAVING

WHERE	HAVING
Filters rows	Filters groups
Used before GROUP BY	Used after GROUP BY
Cannot use aggregate functions	Can use aggregate functions

What is Denormalization?

Combining tables to improve performance is called denormalization

JOINS in SQL

"JOIN is used to combine data from two or more tables based on a same column."

JOIN in SQL

"JOIN is used to combine data from two or more tables using a common column."

---

1. INNER JOIN

- Returns only matching rows from both tables
- Non-matching rows are not shown

```
SELECT *
FROM Student
INNER JOIN Course
ON Student.ID = Course.ID;
```

✅ Only common data

---

2. LEFT JOIN

- Returns all rows from left table
- Matching rows from right table
- If no match, NULL values come from right table

```
SELECT *
FROM Student
LEFT JOIN Course
ON Student.ID = Course.ID;
```

✅ All left table rows

---

3. RIGHT JOIN

- Returns all rows from right table
- Matching rows from left table
- If no match, NULL values come from left table

```
SELECT *
FROM Student
RIGHT JOIN Course
ON Student.ID = Course.ID;
```

✅ All right table rows

---

4. FULL JOIN

- Returns all matching and non-matching rows from both tables
- If no match, NULL values are displayed

```
SELECT *
FROM Student
FULL OUTER JOIN Course
ON Student.ID = Course.ID;
```

✅ All rows from both tables

5. CROSS JOIN

- Returns every possible combination of rows
- every row of first table combines with every row of second table."
- Returns Cartesian product of tables

```
SELECT *
FROM Student
CROSS JOIN Course;
```

6. NATURAL JOIN

- "NATURAL JOIN automatically joins tables based on columns having same names and same data types."No need to write ON condition

```
SELECT *
FROM Student
NATURAL JOIN Course;
```

SELF JOIN

SELF JOIN is a join where a table is joined with itself.”

It is used when rows in the same table are related to each other.

SELF JOIN means a table is joined with itself to compare rows within the same table.”

What is DISTINCT?

DISTINCT removes duplicate values.

```
SELECT DISTINCT department
FROM Employee;
```

What is NULL?

NULL means missing or unknown value.

Constraints in SQL

“Constraints are rules applied on table columns to restrict invalid data and maintain data accuracy and integrity.”

Types of Constraints

1. NOT NULL

- Column cannot contain NULL values

```
Name VARCHAR(20) NOT NULL
```

✅ Value must be entered

---

2. UNIQUE

- Duplicate values are not allowed

```
Email VARCHAR(30) UNIQUE
```

✅ Only unique values allowed

---

3. PRIMARY KEY

- Uniquely identifies each row
- Cannot contain NULL or duplicate values

```
ID INT PRIMARY KEY
```

✅ Unique + Not Null

---

4. FOREIGN KEY

- Creates relationship between tables
- Refers to primary key of another table

```
FOREIGN KEY (Dept_ID)
REFERENCES Department(Dept_ID)
```

✅ Links two tables

---

5. CHECK

- Restricts values based on condition

```
Age INT CHECK (Age >= 18)
```

✅ Condition must satisfy

---

6. DEFAULT

- Assigns default value if no value is given

```
City VARCHAR(20) DEFAULT 'Hyderabad'
```

✅ Automatic default value

GROUP BY

GROUP BY is used to group rows having the same values in a column and is mainly used with aggregate functions like COUNT, SUM, AVG, MAX, and MIN.”

View

“A VIEW is a virtual table created from one or more tables using a SQL query.”

- It does not store data physically
- It shows data from original tables

Difference Between UNION and UNION ALL

UNION	UNION ALL
Removes duplicates	Keeps duplicates
Slower	Faster

What is a Cursor?

Cursor processes records row by row.

Difference Between COUNT(*) and COUNT(column)

COUNT(*)	COUNT(column)
Counts all rows	Ignores NULL

What is the Difference Between SQL and MySQL?

SQL	MySQL
Query language	Database software
Used to write queries	Used to store/manage database

What is DBMS?

DBMS (Database Management System) is software used to store and manage data.

Examples:

- MySQL
- Oracle Database
- Microsoft SQL Server

Stored Procedure

- A Stored Procedure is a set of precompiled SQL statements stored in the database that can be executed repeatedly whenever required."

Trigger

A trigger is a block of code that automatically executes whenever an event occurs on a table."

A trigger is a special type of stored procedure

Difference Between Trigger and Stored Procedure

Trigger	Stored Procedure
Executes automatically	Executes manually
Runs on INSERT, UPDATE, DELETE	Called using `CALL` or `EXEC`
Cannot be called directly	Can be called directly
Associated with a table	Independent database object

Index

An index is used for fast data retrieval and to improve the performance of SQL queries."

Alternate Key

Candidate keys that are not selected as primary key are called alternate keys.

Difference Between Primary Key and Unique Key

Primary Key	Unique Key
Cannot contain NULL	Can contain one NULL
Only one per table	Multiple allowed
Uniquely identifies record	Prevents duplicate values

candidate key

A candidate key is a minimal super key used to uniquely identify each record in a table."

A candidate key is a column or combination of columns that uniquely identifies each record in a table with minimum attributes."

All candidate keys are super keys, but all super keys are not candidate keys."

composite key

"A composite key is a key formed by combining two or more columns to uniquely identify each record in a table."

Simple Understanding

Sometimes a single column cannot uniquely identify records.
So we combine multiple columns.

That combination is called a composite key.

When a primary key is formed using multiple columns, it is called a composite key."

PRIMARY KEY(student_id, course_id)

Aggregate Functions

Definition

"Aggregate functions perform calculations on multiple rows and return a single value."

Functions

- COUNT()
- SUM()
- AVG()
- MAX()
- MIN()

IN Operator

Definition

"IN is used to match multiple values in a WHERE condition."

Example:

```
SELECT *
FROM Student
WHERE city IN ('Hyderabad', 'Chennai');
```

CASE Statement

Definition

"CASE is used for conditional logic in SQL."

Example:

```
SELECT name,
CASE
   WHEN marks >= 90 THEN 'Excellent'
   ELSE 'Good'
END AS Result
FROM Student;
```

ANY Operator

"ANY operator returns true if at least one condition is satisfied."

```
SELECT Name
FROM Student
WHERE Marks > ANY (40, 60);
```

✅ True if condition matches at least one value

ALL Operator

“ALL operator returns true only if all conditions are satisfied.”

```
SELECT Name
FROM Student
WHERE Marks > ALL (40, 60);
```

✅ True only if condition matches all values

What is Schema?

Definition

“Schema is the logical structure of a database containing tables, views, relationships, etc.”

Entity Integrity & Referential Integrity

Entity Integrity

Primary key cannot contain NULL.

Referential Integrity

Foreign key value must exist in referenced table.

AUTO_INCREMENT

Definition

“AUTO_INCREMENT automatically generates unique numbers.”

Example:

```
id INT AUTO_INCREMENT PRIMARY KEY
```

LIKE Operator

Definition

"LIKE is used for pattern matching."

Example:

```
SELECT *
FROM Student
WHERE name LIKE 'R%';
```

`R%` → names starting with R

ACID Properties (Very Important)

Definition

"ACID properties ensure reliable transaction management in databases."

transaction is treated as a single unit

Properties

- **Atomicity** → Transaction happens completely or not at all
- **Consistency** → Database remains valid
- **Isolation** → Transactions do not affect each other
- **Durability** → Once transaction is committed, changes are permanently saved even if system crashes."
- # What is DBMS vs RDBMS?

DBMS	RDBMS
Stores data	Stores related data
No relationships	Relationships maintained
Less security	More security

- "RDBMS is an advanced form of DBMS that stores related data using tables."

What is a Transaction?

"A transaction is a sequence of SQL operations performed as a single unit."

Example:

- Bank money transfer

What is UNION?

“UNION combines results of two queries and removes duplicates.”

What is Alias?

“Alias is a temporary name given to a table or column.”

Example:

```
SELECT Name AS Student_Name
FROM Student;
```